

MICROBIOLOGY AND FOOD SAFETY

Efficacy of feed additives against *Campylobacter* in live broilers during the entire rearing period: Part B

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ABSTRACT A total of 636 day-of-hatch Ross 308 broilers chicks were used in 4 independent trials carried out to screen the effect of 12 feed additives on reducing cecal colonization of *Campylobacter* in broilers. The tested additives were probiotics based on *B. subtilis* and *S. cerevisiae*, a garlic extract, a blend of herbal substances and essential oils, two different combinations of essential oils and organic acids (OA), two mixtures of flavoring compounds, medium chain fatty acids (MCFA), monoglycerides (MG) of MCFA and MG-MCFA+OA. At 14 days of age, all the birds were orally infected with 0.1 mL of a bacterial suspension of *C. jejuni* ST-45 diluted at 10⁵ cfu/mL in tryptone salt broth. In each trial, there was a positive control group and 2 (Trials 1 and 2) or 4 (Trials 3 and 4) additional treatment groups supplemented with additives, which were added to feed or water only to the finisher (21 to 42 d) diet

(Trials 1 and 2) or to the starter (0 to 21 d) and finisher diets (Trials 3 and 4). Feed and water were available ad libitum. On days 35 and 42 of age in Trials 1 and 2, and on days 21, 35 and 42 of age in Trials 3 and 4, 10 (Trials 1 and 2) or 12 birds (Trials 3 and 4) per group were euthanized for cecal sampling. In Trial 1, birds fed with MCFA and MG-MCFA had a significant ($P < 0.05$) reduction in cecal *Campylobacter* colonization compared to control at 35 d, but only the group treated with MG-MCFA maintained the reduction at 42 d. In Trials 2 to 4, no significant differences ($P > 0.05$) in cecal *Campylobacter* counts were found between the treated and control animals. In conclusion, although none of the treatments were able to completely prevent the colonization of chickens with *C. jejuni*, MCFA and MG-MCFAs could reduce the pathogen counts when supplemented from 21 days onwards.

Key words: *Campylobacter jejuni*, broiler, feed additive, CAMPYBRO

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INTRODUCTION

Campylobacter is the major cause of bacterial gastroenteritis in humans worldwide. In the EU, it is the most commonly reported gastrointestinal bacterial pathogen since 2005, with 214,779 human cases reported in 2013 (EFSA and ECDC, 2015).

The EU baseline survey in the broiler meat chain carried out in 2008 revealed that the mean prevalence of *Campylobacter* is respectively 71.2% and 75.8% for broiler batches (cecal content) and carcasses (EFSA, 2010), although it varies widely from 2.0% to 100.0% between the Member States (EFSA, 2010). As a consequence, the contaminated broiler meat products may serve as a source for cross-contamination to other food-

stuffs and surfaces during meal preparation in the consumer's kitchen (Luber et al. 2006; Fravallo et al., 2009). Therefore, implementation of *Campylobacter* control measures at the primary production level are needed for a reduction in human exposure.

Few studies have evidenced a positive role of probiotics in preventing or reducing the shedding of *C. jejuni* in in vivo trials (Morishita et al., 1997; Willis and Reid, 2008; Santini et al., 2010), although in vitro studies reported a strong antimicrobial activity of several probiotic strains toward this pathogen (Chang and Chen, 2000; Fooks and Gibson, 2002; Chaveerach et al., 2004). A wide range of plant extracts and compounds have demonstrated strong bactericidal activity against *Campylobacter* spp. in vitro (Friedman et al., 2002; Hermans et al., 2011; Sirirak and Voravuthikunchai, 2011; Robyn et al., 2013). However, when evaluating in vivo results, data are not so conclusive. Also, in vitro studies have also demonstrated that organic acids (OA), medium chain fatty acids (MCFA) or

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monoglycerides of MCFA (**MG-MCFA**) have a strong bactericidal effect on *Campylobacter* spp. (Chaveerach et al., 2002; Hermans et al., 2010). However, inconsistent results have been reported in in vivo trials (Van Deun et al., 2008; Solis de los Santos et al., 2008a,b, 2010; Molatová et al., 2011; Hermans et al. 2012).

It is clear that there exist differences in the results both in vitro and in vivo. Therefore, the objective of this work was to screen the effect of 24 feed additives of several families on reducing cecal colonization of *Campylobacter* in broilers. Because of the large number of treatment groups, the study was divided into two parts [A (Guyard-Nicodème et al., 2015) & B (current study)] to obtain more representative results, each part evaluates 12 feed additives.

MATERIAL AND METHODS

Experimental Design

Four independent trials were carried out to test the 12 products. In Trial 1, 36 day-of-hatch Ross 308 broilers chicks (50% male and 50% female) were used per treatment group. All birds received the same common basal diet until 21 days. Three treatments were applied to the feed from 21 to 42 days: T1 Control, T2 MCFA (C8+C10) and T3 MG-MCFA (C8+C10). In Trial 2, 36 day-of-hatch Ross 308 broilers chicks (50% male and 50% female) were used per treatment group. All birds received the same common basal diet until 21 days. Three treatments were applied to the feed from 21 to 42 days: T1 Control, T2 garlic-based extract and T3 *Bacillus subtilis* DSM 17299. In Trial 3, 42 day-of-hatch Ross 308 broilers chicks (50% male and 50% female) were used per treatment group. Five treatments were applied to the feed from 1 to 42 days: T1 Control, T2 essential oil components including thymol, eugenol and piperine and benzoic acid, T3 Herbal substances and an essential oils mixture (containing thymol and anethole, spices and saponins), T4 *Saccaromyces cerevisiae boulardii* and T5 Propionic acid glycerol esters and oregano. In Trial 4, 42 day-of-hatch Ross 308 broilers chicks (50% male and 50% female) were used per treatment group. Five treatments were applied from 1 to 42 days: T1 Control, T2 MG of C3-C12 in water, T3 MG of C3-C12 + OA in water, T4 Ca propionate, sorbic acid and flavoring compounds and T5 coated source of formic acid, citric acid and an encapsulated blend of essential oils including thyme, oregano, capsicum and citrus extract. A summary of the experimental treatments applied in each experiment is included in Table 1. The trials were carried out in a facility with animal biosafety Level 2. The birds were kept in cages in groups of three birds with an area of 0.21 m² (0.50 m × 0.42 m). The building was supplied with artificial, programmable lights 24 h/d, automated electric heating and forced ventilation.

The experimental diets consisted of standard non-medicated cereal (corn/wheat/barley)-soy based diets. The starter diet was offered to birds from 1-day-old

until 21 days of age and the finisher diet from day 22 to 42 days. Feeds were presented as mash. Feed and water were available ad libitum.

Husbandry, euthanasia methods, experimental procedures and biosafety precautions were approved by the Ethical Committee of the Faculty of Veterinary Medicine, Murcia University.

Broiler weights and feed intake were determined per cage at 21, 35 and 42 days of age.

Broiler Inoculation

All the animals were individually and orally infected at 14 days of age with 0.1 mL of a bacterial suspension of *Campylobacter jejuni* ST-45 diluted at 10⁵ cfu/mL in tryptone salt broth.

The strain *C. jejuni* ST-45 was isolated from Spanish broiler flocks and was stored at -70°C in peptone broth containing 20% (vol/vol) glycerol in Laboratorio Central de Veterinaria (Algete, Spain). Five days before each inoculation, the strain was recovered from frozen stock after plating on mCCDA (selective modified Charcoal Cefoperazone Deoxycholate Agar) at 42°C for 48 h under a microaerophilic atmosphere (85% N₂, 10% CO₂ and 5% O₂). One colony was then used to inoculate 10 mL Brucella broth. After 24 h at 42°C under a microaerophilic atmosphere the bacterial suspension was diluted at 10⁵ cfu/mL in tryptone salt broth.

Sampling and Microbiological Analysis

Prior to oral infection with *C. jejuni*, two broiler chicks per experimental treatment were randomly selected and euthanized in order to check the absence of *C. jejuni* infection in the ceca. Analyses were carried out according to the ISO 10272 standard. All broilers tested negative. The concentration of *Campylobacter* in the administered inoculum was also determined after the oral infection in each trial. On days 35 and 42 of age in Trials 1 and 2 and on days 21, 35 and 42 of age (corresponded to 7, 21 and 28 days post challenge, respectively) in Trials 3 and 4, ten (Trials 1 and 2) or 12 birds (Trials 3 and 4) from each treatment group were euthanized for cecal sampling. *Campylobacter* was recovered from caeca using direct plating and enumeration. Cecal contents were homogenized and 1 g diluted 1/10 (wt/vol) in tryptone salt broth. After homogenization, enumeration was carried out by serial dilution in tryptone salt broth in order to assess *C. jejuni* count on mCCDA plates after 44 ± 4 h of incubation at 42 ± 1°C in a microaerophilic atmosphere (85% N₂, 10% CO₂ and 5% O₂). The detection limit for enumeration of the *Campylobacter* was 1 × 10² cfu/g of cecal content.

Statistical Analysis

Data were analyzed by IBM SPSS 19.0 Statistics for Windows Version 19.0. (IBM Corp., Armonk, NY). The

Table 1. Tested products and doses.

Trial	Treatment Composition	Dose ^a
Trial 1	T1 Control	–
	T2	C8 and C10 mixture 50%
	T3	Monoglycerides of C8 and C10 50%
Trial 2	T1 Control	–
	T2	Garlic based extract
	T3	<i>Bacillus subtilis</i> DSM 17299
Trial 3	T1 Control	–
	T2	Essential oil components including thymol, eugenol and piperine + benzoic acid
	T3	Herbal substances and essential oils mixture (containing thymol and anethole, spices and saponins)
	T4	<i>Saccharomyces cerevisiae boulardii</i>
	T5	Propionic acid glycerol esters and oregano
Trial 4	T1 Control	–
	T2 (in water)	Monoglycerides of C3, C4, C8, C10 and C12
		0.500% (vol/vol) starter
		0.250% (vol/vol) finisher
	T3 (in water)	Monoglycerides of C3, C4, C8, C10 and C12, formic acid, ammonium formate, sodium formate and sodium sorbate
		1.000% (vol/vol) starter
		0.500% (vol/vol) finisher
	T4	Calcium propionate, sorbic acid and flavoring compounds
		0.300% (wt/wt)
	T5	Coated source of formic acid and citric acid and encapsulated blend of essential oils including thyme, oregano, capsicum and citrus extract
		0.500% (wt/wt)

^aProducts were added at the same dose in the two feeds, unless specified. Starter diet: 0 to 21 d, finisher diet: 21 to 42 d.

individual logarithms of the 10 to 12 birds per group either at 21, 35 or 42 days of age of each experiment were used as the experimental unit for statistical analysis. The Shapiro-Wilk test was used to test the normal distribution of the data. When the distribution of data was normal, data were analyzed as a completely randomized design through analysis of variance (ANOVA) followed by the Tukey's test to find the significance between groups. The nonparametric test of Kruskal-Wallis was used when the criteria of normality and homogeneity of the variances were not validated, followed by the Dunn's test to find the significance between groups. Statistical significance was declared at a *P*-value of ≤ 0.05 , with $0.05 < P \leq 0.10$ considered as a near-significant trend.

RESULTS

The bacteriological analysis of cecal samples collected from the broilers before *C. jejuni* challenge demonstrated that the broiler chicks did not naturally contained cecal *Campylobacter* spp. Body weights and feed consumption were normal and were not affected ($P < 0.05$) by the supplementation of any of the additives tested when compared with control treatment in any of the trials (data not shown).

Campylobacter jejuni colonization in Trial 1

Broiler chicks in this experiment received a common diet until 7 d post challenge, when they started to eat their respective experimental diets. The anal-

ysis of cecal samples collected from 15 broiler chicks at 7 d post challenge demonstrated a *Campylobacter* cecal colonization of $8.68 \pm 1.27 \log_{10}$ cfu/g. At 21 d post challenge (35 d of age), the control group had a mean colonization level of $8.66 \log_{10}$ cfu/g, but treated groups with MCFA and MG-MCFA showed a significant ($P < 0.05$) reduction in colonization (7.75 and $7.72 \log_{10}$ cfu/g, for MCFA and MG-MCFA groups, respectively). There was no significant difference between both treated groups (Table 2). At 28 d after challenge (42 d of age), the group treated with MG-MCFA also showed a significant reduction in *Campylobacter* colonization compared to the control and the MCFA groups (0.88 and $0.81 \log$ reduction, respectively; $P < 0.05$). No significant difference between the control and the MCFA group was observed.

Campylobacter jejuni colonization in Trial 2

Broiler chicks in this experiment received a common diet until 7 d post challenge, when they started to eat their respective experimental diets. The analysis of cecal samples collected from 15 broiler chicks at 7 d post challenge, demonstrated a *Campylobacter* cecal colonization of $7.95 \pm 0.96 \log_{10}$ cfu/g. The results showed no significant differences ($P > 0.05$) in cecal *Campylobacter* counts of broilers receiving control, garlic extract or *B. subtilis* supplemented diets at 21 or 28 d post challenge (Table 3).

Table 2. Effect of dietary treatment on the cecal colonization of *Campylobacter jejuni* in broiler chickens in Trial 1.¹

Treatment	7 d post challenge	21 d post challenge	28 d post challenge
1 Control		8.66 ^a	8.40 ^a
2 C8 and C10	8.68 ± 1.26 ²	7.75 ^b	8.33 ^a
3 Monoglycerides of C8 and C10		7.72 ^b	7.52 ^b
SEM	–	0.256	0.206
Probability	–	0.0268	0.0242

^{a,b}Means within the same column with different superscripts are significantly different ($P < 0.05$).

¹Data presented as means of logarithms of 10 cecal samples per group ($\log_{10}$ cfu/g).

²Mean ± SD from 15 birds selected at random (5 per treatment). All the animals had received the same control diet from 0 to 21 days without additives.

Table 3. Effect of dietary treatment on the cecal colonization of *Campylobacter jejuni* in broiler chickens in Trial 2.¹

Treatment	7 d post challenge	21 d post challenge	28 d post challenge
1 Control		7.36	6.30
2 Garlic based extract	7.95 ± 0.96 ²	7.66	6.05
3 <i>Bacillus subtilis</i> DSM 17299		7.54	6.72
SEM	–	0.282	0.365
Probability	–	0.7530	0.4350

¹Data presented as means of logarithms of 10 cecal samples per group ($\log_{10}$ cfu/g).

²Mean ± SD from 15 birds selected at random (5 per treatment). All the animals had received the same control diet from 0 to 21 days without additives.

Table 4. Effect of dietary treatment on the cecal colonization of *Campylobacter jejuni* in broiler chickens in Trial 3.¹

Treatment	7 d post challenge	21 d post challenge	28 d post challenge
1 Control	7.40	7.38	7.71
2 Essential oils (incl. thymol, eugenol and piperine) + Benzoic acid	7.80	7.26	7.65
3 Herbal substances + essential oils (containing thymol and anethole, spices and saponins)	7.43	7.77	6.91
4 <i>S. cerevisiae boulardii</i>	7.77	7.13	7.44
5 Propionic acid glycerol esters and oregano	7.32	7.03	7.83
SEM	0.352	0.292	0.332
Probability	0.8010	0.4360	0.3230

¹Data presented as means of logarithms of 12 cecal samples per group ($\log_{10}$ cfu/g).

Campylobacter jejuni colonization in Trial 3

In this trial, all the dietary treatments were supplemented during the entire trial period. None of the treatments were able to prevent the colonization with *Campylobacter* at 7 d post challenge (Table 4). The results showed no significant differences ($P > 0.05$) in cecal *Campylobacter* counts of broilers receiving control, a blend of herbal substances and essential oils, two different combinations of essential oils and organic acids or *S. cerevisiae* supplemented diets at 21 or 28 d post challenge.

Campylobacter jejuni colonization in Trial 4

In this trial, all the treatments were also supplemented during the entire trial period. Two treatments,

MG-MCFA and MG-MCFA with a mixture of OA, were administered through the drinking water, and other two treatments were supplemented to the feed. None of the treatments was able to prevent the colonization with *Campylobacter* at 7 d post challenge (Table 5). The results showed no significant differences ($P > 0.05$) in cecal *Campylobacter* counts of broilers receiving control, MG-MCFA or MG-MCFA+OA in water, a mixture of flavoring compounds or a blend of flavoring compounds and OA supplemented diets at 21 or 28 d post challenge.

DISCUSSION

The aim of this research was to search for feed additives able to reduce cecal colonization of *Campylobacter* in broilers at slaughter age in vivo. This was planned in order to avoid the discrepancies previously

Table 5. Effect of dietary treatment on the cecal colonization of *Campylobacter jejuni* in broiler chickens in Trial 4.¹

Treatment	7 d post challenge	21 d post challenge	28 d post challenge
1 Control	7.52	7.66	7.53
2 Monoglycerides of C3, C4, C8, C10 and C12	6.82	7.22	6.87
3 Monoglycerides of C3, C4, C8, C10 and C12 + Organic acids	6.62	5.36	5.80
4 Calcium propionate, sorbic acid, flavoring compounds	7.42	8.48	8.07
5 Coated formic + citric acids & encapsulated blend of essential oils (incl. thyme, oregano, capsicum and citrus extract)	7.93	7.51	6.81
SEM	0.382	0.691	0.747
Probability	0.1670	0.2950	0.6450

¹Data presented as means of logarithms of 12 cecal samples per group (log₁₀ cfu/g).

found between the results in vitro and in vivo (Van Deun et al., 2008; Hermans et al., 2011; Robyn et al., 2013). For that purpose, additives belonging to different families of compounds were tested. None of the two probiotics assayed (*B. subtilis* or *S. cerevisiae*) have shown significant reduction in *C. jejuni* colonization. Maruta et al. (1996) evaluated different administration periods of *Bacillus subtilis* C-3102 and slaughter ages after infection and concluded that differences were only significant at the end of the experiment, after a continuous feeding period. In the present trial, *B. subtilis* was supplemented only from 21 to 42 d, which possibly impaired the effectiveness of the probiotic. Our results with *S. saccharomyces* are in agreement with the findings of Line et al. (1998), which found that *Campylobacter* colonization of broilers was not significantly affected by live yeast treatment. Other probiotics have also been effective against *Campylobacter*. For example, Santini et al. (2010) reported that *C. jejuni* concentration in poultry feces was significantly reduced after 15 consecutive days of administration of *Bifidobacterium longum* PCB 133. Willis and Reid (2008) showed a lower level of *C. jejuni* in broiler chickens fed with a standard diet supplemented with a mixed probiotics containing *Lactobacillus acidophilus*, *Lactobacillus casei*, *Bifidobacterium thermophilus* and *Enterococcus faecium*. Interestingly, a 6-log significant reduction was observed by Ghareeb et al. (2012) after receiving for 8 or 15 d a multispecies probiotic product containing *Enterococcus faecium*, *Pediococcus acidilactici*, *Bifidobacterium animalis*, *Lactobacillus salivarius* and *Lactobacillus reuteri* strains added to drinking water. Although probiotic treated broilers seem to have lower levels of *Campylobacter*, Willis and Reid (2008) suggested that direct-fed microbials alone might not be sufficient to reduce *Campylobacter* presence to acceptable levels. Therefore, the bacterial species, time of administration and synergies with other additives are key factors in future research in this field.

Two products containing plant extracts and 3 products containing a mixture of plant extracts and organic

acids have been evaluated without any inhibitory effect against the colonization of *C. jejuni* in broilers. Several plant extracts and compounds have been found to exert strong bactericidal activity against *Campylobacter* spp. in vitro (Friedman et al., 2002; Hermans et al., 2011; Sirirak and Voravuthikunchai, 2011; Robyn et al., 2013). However, when evaluating in vivo results, data are not so conclusive. In this respect, the results obtained in the present work are in line with those from Hermans et al. (2011), that showed that dietary supplementation with trans-cinnamaldehyde had no protective effect on chicks infected with *C. jejuni*. Similarly, Kurekci et al. (2014) found that the mean log₁₀ counts of *C. jejuni* were similar across treatments after feeding broilers during 7 wk with a diet with plant extracts (*Acacia decurrens*, *Eremophila glabra*), the essential oil lemon myrtle or plant secondary compounds such as terpinene-4-ol. Also, Robyn et al. (2013) observed that broilers provided with allicin-containing drinking water did not have significantly lower cecal *C. jejuni* numbers compared to broilers in the control groups, although a trend of almost 1 log₁₀ toward lower cecal *C. jejuni* numbers could be observed. However, Peinado et al. (2012) showed that dietary supplementation with the garlic-derived product propyl propane thiosulfonate had a significant antimicrobial effect in vivo against *C. jejuni* in the ileal content.

In vitro studies have also demonstrated that organic acids have a strong bactericidal effect on *Campylobacter* spp. (Chaveerach et al., 2002). For example, a strong in vitro bactericidal effect of butyrate on *C. jejuni* was observed by Van Deun et al. (2008). However, when butyrate-coated micro-beads was supplemented to broiler feed, it was not able to decrease the final cecal *C. jejuni* count at the end of the experiment. Skånseng et al. (2010) demonstrated that chickens offered feed supplemented with a combination of formic acid at a high dose and sorbate were protected against *C. jejuni* colonization during different infection trials.

The discrepancies between the in vivo and in vitro results might be partly explained by the absorption of

the active compounds in the upper part of the gastrointestinal tract, by its degradation by the digestive enzymes, by the interaction of the active substances with other intestinal/cecal constituents or bacteria or by the presence of mucus in the intestinal and cecal environment of broilers. In fact, Robyn et al. (2013), Hermans et al. (2010) and Van Deun et al. (2008) have already proven that chicken intestinal mucus protects *C. jejuni* from the antibacterial activity of allicin, capric acid and sodium butyrate.

A product containing MCFA, two products containing a blend of MG-MCFA and a product containing MG-MCFA and OA have been evaluated, with a significant reduction in the colonization of *C. jejuni* in broilers when MCFA were supplemented to the feed, especially as MG. Solis de los Santos et al. (2008b) reported the therapeutic effect of the supplementation of caprylic acid in feed for the last 3 days, since it reduced *Campylobacter* populations in the cecal content of 15 days infected chicks by 3 to 4 logs in 3 separate trials. Also, caprylic acid consistently reduced *Campylobacter* counts in 10-day-old chicks when supplemented in-feed during all the experimental period in 4 separate trials (Solis de los Santos et al., 2008a). In another study, caprylic acid supplemented in feed for 3 or 7 days before slaughter reduced *Campylobacter* compared with the positive controls (Solis de los Santos et al., 2009, 2010). Similarly, the treatment of *Campylobacter*-infected broilers by adding monocaprin emulsions to their drinking water and feed significantly reduced the number of *Campylobacter* in cloacal swabs (Hilmarsen et al., 2006). Also, Van Gerwe et al. (2010) concluded that the number of *C. jejuni* bacteria required to colonize 50% of broilers was estimated to be 200 times higher when the diets were supplemented with caprylic, capric and lauric acids. Metcalf et al. (2011) found that birds dosed with 0.175% caprylic acid had a significant reduction in cecal *Campylobacter* counts when compared with the positive control. Similarly, Hermans et al. (2010) concluded that in-feed MCFA caproic, caprylic and capric acids, or their sodium salts, from 3 d before euthanization, were not capable of reducing cecal *Campylobacter* colonization in 27-day-old broilers experimentally infected with *C. jejuni* at 15 d of age. Similar results were obtained by Hermans et al. (2012) and no significant differences in cecal *Campylobacter* counts of broilers receiving an emulsion of a mixture of caproic, caprylic, capric and lauric acids in the drinking water. Our data showed effective results of MCFA and MG-MCFA reducing *Campylobacter* counts when supplemented in-feed, but no significant results were observed when included into the drinking water. It can be hypothesized that addition of MCFA to the drinking water could be less effective and produced inconsistent results, at least with the dose used in this experiment. Thus, Molatová et al. (2011) found that supplementing broiler feeds with a coated or non-coated mixture of caprylic and capric acids significantly decreased fecal *C. jejuni* counts by 1.3 to 4.1 log₁₀ cfu/g

four days after inoculation. They also concluded that the encapsulation enhanced the efficacy of the acids.

In conclusion, none of the treatments completely prevented the colonization of chickens with *C. jejuni*, and only MCFA at 35 d or MG-MCFA at 35 and 42 d significantly decreased its cecal population. Probably several strategies will have to be combined in order to find a suitable, reliable and effective measure to control *Campylobacter* in poultry flocks. Moreover, large-scale field trials will need to be performed to determine the practical effect of such applications in commercial poultry conditions.

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